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CAPSTONE PROJECT

SATELLITE INTERNET COMMUNICATION USING OSI LAYERS

CSA0711-COMPUTER NETWORKS

UNDER THE SUPERVISION OF:
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INTRODUCTION

- Satellite Internet Communication provides internet access through communication satellites.
- It connects remote and rural areas where wired networks are unavailable.
- The OSI model divides network communication into seven functional layers.
- Each OSI layer performs a specific role in data transmission.
- Data travels from the user device to the satellite and then to the destination server.





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OBJECTIVES

- Understand the fundamentals of satellite internet communication.
- Explain the functions of all seven OSI layers.
- Demonstrate data transmission through satellite networks.
- Analyze uplink and downlink communication.
- Understand data encapsulation and decapsulation.
- Study the role of networking protocols in satellite communication.
- Identify challenges such as latency and signal interference.





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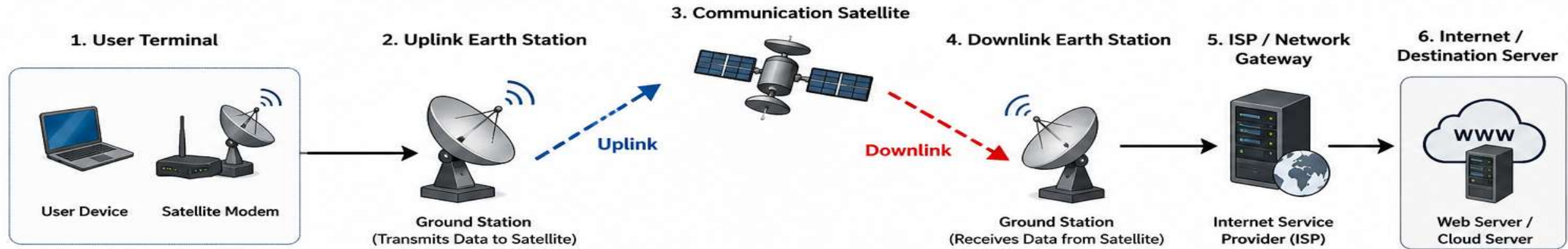
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PROBLEM STATEMENT

- Providing reliable internet access in remote and rural areas is challenging.
- Traditional wired networks are costly and difficult to deploy in such regions.
- Satellite communication experiences high latency and signal interference.
- Efficient data transmission requires proper coordination of all OSI layers.
- Understanding how the OSI model supports satellite communication can improve network performance



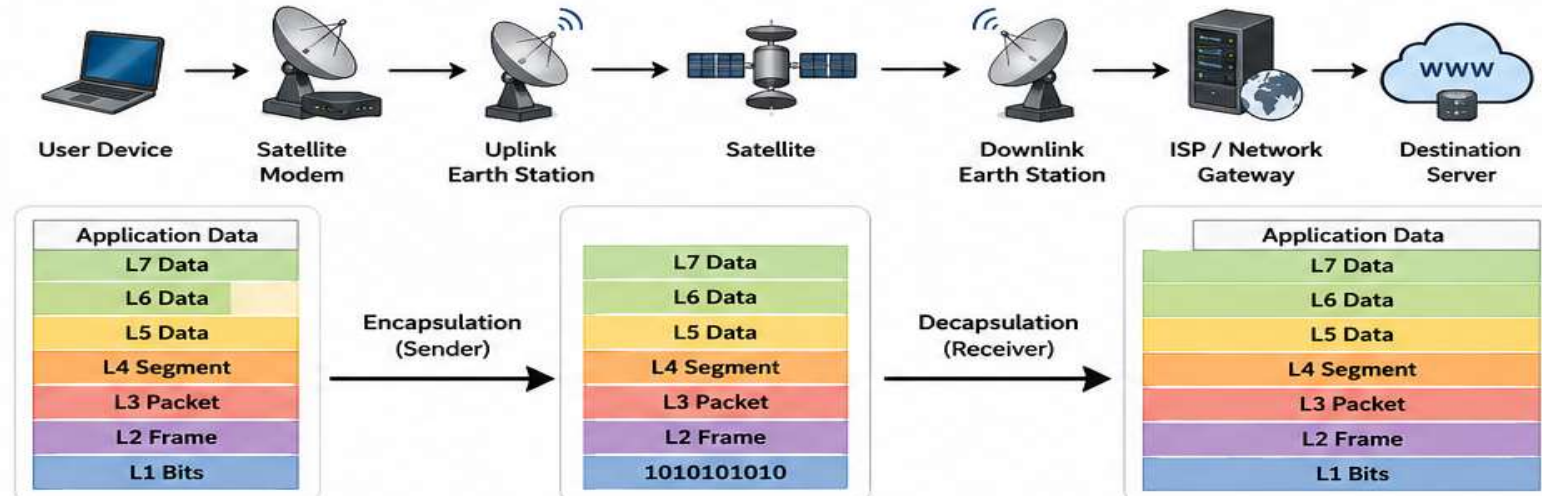
SATELLITE INTERNET COMMUNICATION USING OSI LAYERS SYSTEM ARCHITECTURE



OSI MODEL

7	Application Layer	Provides network services
6	Presentation Layer	Data format, encryption, compression
5	Session Layer	Session establishment, management, termination
4	Transport Layer	End-to-end communication, reliability (TCP/UDP)
3	Network Layer	Logical addressing and routing (IP)
2	Data Link Layer	Framing, error detection, flow control
1	Physical Layer	Transmission of bits as radio signals

DATA FLOW





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OSI LAYER-BASED SATELLITE COMMUNICATION SYSTEM

- **Application Layer** – Provides services such as web browsing, email, and video streaming.
- **Presentation Layer** – Formats, encrypts, and compresses data before transmission.
- **Session Layer** – Establishes, manages, and terminates communication sessions.
- **Transport Layer** – Ensures reliable data delivery using TCP or fast transmission using UDP.
- **Network Layer** – Handles IP addressing and routes data packets between networks.
- **Data Link Layer** – Performs framing, error detection, and communication between satellite devices.
- **Physical Layer** – Transmits data as radio frequency (RF) signals through the satellite communication channel.
- **Data Flow** – User Device → Satellite Modem → Ground Station → Satellite → ISP → Destination Server.



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END-TO-END SATELLITE INTERNET PROTOCOL

- User sends a data request from a device (computer/mobile).
- The satellite modem converts the data into RF signals.
- Data is transmitted to the satellite through the **uplink**.
- The satellite relays the signal to the ground station via the **downlink**.
- The ISP gateway routes the data to the destination server using IP protocols.





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SIMULATING SATELLITE LINK PROTOCOL LAYER

- Simulates data transmission through a satellite communication link.
- Demonstrates packet flow across all seven OSI layers.
- Shows data encapsulation before transmission and decapsulation after reception.
- Simulates uplink and downlink communication between ground stations and satellites.
- Illustrates routing, error detection, and data delivery processes.
- Analyzes transmission delay (latency) and signal propagation.
- Verifies reliable communication using network protocols.
- Helps visualize the working of satellite communication in a real-world network.



ACHIEVED LATENCY AND THROUGHPUT IMPROVEMENTS

This table presents typical performance metrics and operational considerations across different OSI layers within satellite internet communication systems.

OSI Layer	Performance Metric	Typical Value	Scenario/Context
Physical	Round-Trip Delay (RTD)	580 ms	GEO Satellite Link
Data Link	Effective Throughput	2.5 Gbps	LEO User Terminal Aggregate
Network	Packet Error Rate (PER)	0.05 %	Ku-band Satellite Uplink
Transport	Recommended TCP Window Size	2 MB	High Latency (GEO) Adaptation



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CONCLUSION

- Demonstrates how the OSI model supports satellite internet communication.
- Explains the role of each OSI layer in reliable data transmission.
- Shows end-to-end communication from the user to the destination server.
- Highlights uplink, downlink, and satellite data routing processes.
- Improves understanding of real-world satellite networking concepts.
- Emphasizes reliable, secure, and efficient internet communication through satellites.
- Useful for learning computer networks and communication systems.



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THANK YOU